Vidit Gupta

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Education

Dwarkadas J. Sanghvi College of Engineering — CGPA: 8.98
Bachelor's of Engineering in Information Technology — *Sep. 2024 – 2028*

Jawahar Navodaya Vidyalaya — 12th Boards: 92%
Secondary and High School — *Jul. 2016 – Apr 2024*

Technical Skills

Programming Languages: Python, C, C++, JavaScript
Deep Learning Frameworks: PyTorch, TensorFlow
Libraries: NumPy, Pandas, Matplotlib, scikit-learn, Weights & Biases
Tools & Platforms: Git, Linux, Jupyter, Google Colab
Research Methods: Mechanistic Interpretability, Geometric Probing, Attention Analysis, Effective Rank, Intrinsic Dimensionality, CKA

Projects

Residual Stream Geometry in GPT-Style Transformers | *Pytorch, Transformers, Compression* — GitHub

- Conducted mechanistic interpretability study on GPT-style transformers ($d_{\text{model}} = 128, 256, 512$) trained on WikiText-2, probing effective rank, stable rank, participation ratio, and intrinsic dimensionality across all layers.
- Discovered capacity utilization converges to $\approx$79% independent of model width; d=128 achieves better validation loss ($\sim$5.40) than larger models despite 4$\times$ fewer parameters, suggesting overparameterization at small scale.
- Identified Layer-1 as dominant transformation site (CKA $\approx$ 0.55 vs. 0.91–0.97 for deeper layers), with deeper layers acting as refinement stages — revealing architecture-level efficiency patterns relevant to layer pruning.
- Measured intrinsic dimensionality growth ($\approx 2 \rightarrow$ 8–14 across layers), confirming token representations lie on a low-dimensional manifold far below ambient dimension, motivating aggressive residual stream compression.

Attention Geometry & Training Dynamics in Transformers | *Initialization, PyTorch* — GitHub

- Instrumented a 4M-parameter GPT-2 transformer with 18 geometric metrics (participation ratio, effective rank, spectral norm, attention entropy), tracking Q/K/V dynamics across 87 training epochs on a 40M-character WikiText-103 subset.
- Discovered opposing pre/post-softmax effective rank trajectories: QK^T rank increases through training while attention weight rank collapses, quantifying softmax as active geometric compression rather than passive normalization.
- Identified a spectral phase transition in W_q across layers 1–3: dominant singular direction energy peaks then redistributes as Frobenius norm grows, with Layer 0 exhibiting anomalous monotonic behavior.
- Revealed asymmetric gradient pressure on Q, K, V projections despite shared parameterization: V norm collapses $\sim$epoch 1, Q norm decreases steadily, K norm increases in deeper layers — documenting role-specific geometry.

Leadership

- **DJS Codestars** – Competitive Programming Technical Member: organized problem sets and conducted sessions for 50+ members
- **GDG DJSCE** – ML Research Member: contributed to reading groups and ML workshops
- **DJ Init.Ai** – ML Research Mentee: pursuing mentored research in deep learning and model analysis

Achievements & Certifications

- **Competitive Programming**: CodeChef – 1690 rated | Codeforces – 1360 rated
- **Regional Student Science Congress (IIT-Gandhinagar)**: Presented a facial landmark-based mouse controller (OpenCV, MediaPipe) at 16
- **Certifications**: CS50X – Harvard University | Machine Learning Specialization – DeepLearning.AI (Coursera) | Algorithms Specialization – Stanford (Coursera)